



AWS First Cloud AI Journey – **Project Plan**

[SKYNET] – [FPT University] – [Online
Platform for Tracking and Predicting Hurricane
Trajectory]
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1 BACKGROUND AND MOTIVATION

1.1 EXECUTIVE SUMMARY

The customer aims to build an interactive, visually engaging online platform that delivers clear and accessible information about hurricanes—similar to a modern, dynamic weather report focused on storm activity. The platform is designed for anyone who wants to follow recent storms, understand how they behave, and explore predicted trajectories through intuitive visualizations.

Currently, the customer relies on manual data updates and on-premises processing, which limits automation, scalability, and the ability to provide timely storm insights. To overcome these limitations, the customer plans to modernize the entire system using a fully serverless architecture on AWS, enabling rapid data handling, real-time updates, and a more responsive user experience.

The main objective of migrating to AWS is to support an automated, scalable, and resilient storm-tracking platform with minimal operational overhead. Key goals include:

- Automating storm data ingestion, processing, and model-powered trajectory forecasting
- Delivering real-time storm activity, alerts, and projected paths in an interactive visual format
- Leveraging AWS serverless technologies to reduce infrastructure maintenance and maximize cost efficiency
- Ensuring global availability, strong data integrity, and consistent performance through AWS-managed services

Below are some use case:

1. **Real-time Storm Monitoring:** Users access the website through Route 53 and CloudFront to view the latest storm data stored in Amazon S3.
2. **Predictive Modeling:** API Gateway routes prediction requests to AWS Lambda, which loads the latest model and generates forecasts based on stored geodesic storm data.
3. **Automated Data Crawling:** An EventBridge rule triggers a Lambda function every Sunday to crawl and update storm datasets from external sources into S3.
4. **Secure Operations and Deployment:** AWS Secrets Manager stores API keys and credentials, while AWS CodePipeline and CodeBuild automate deployment from GitHub to AWS services.

To achieve the customer's objectives, the AWS Partner will deliver the following professional services:

- **Solution Architecture & Design:** Define and implement a serverless AWS architecture using Lambda, S3, API Gateway, CloudFront, and Route 53.
- **Automation & Data Integration:** Develop and configure EventBridge-triggered Lambda workflows for automated data ingestion and model updates.
- **Model Hosting & Inference:** Deploy and integrate storm prediction models stored in S3 for low-latency inference through API Gateway.

- **Security Implementation:** Configure IAM policies, Secrets Manager, and best practices for access control and credential management.
- **CI/CD Pipeline Setup:** Build a continuous deployment pipeline using CodePipeline and CodeBuild for seamless code updates from GitHub.
- **Testing & Optimization:** Conduct performance, security, and reliability tests to ensure scalability and fault tolerance.

1.2 PROJECT SUCCESS CRITERIA

Success will be measured by achieving the following quantifiable outcomes:

- **Automation:** 100% of storm data updates automatically triggered and processed through EventBridge and Lambda.
- **Performance:** API Gateway endpoints respond to user requests within <500 milliseconds (p95 latency).
- **Availability:** System uptime maintained at or above 99.9% via Route 53 and CloudFront.
- **Security:** All sensitive credentials stored in AWS Secrets Manager with no unauthorized access incidents.
- **Deployment Efficiency:** Code changes automatically built and deployed from GitHub to AWS within 10 minutes.
- **Scalability:** System capable of handling at least 1,000 concurrent user requests without degradation.
- **Cost Optimization:** Achieve at least a 40% cost reduction compared to traditional EC2-based infrastructure.

1.3 ASSUMPTIONS

- The customer will provide access to existing storm datasets and trained predictive models.
- External meteorological data sources remain stable and accessible for automated crawling.
- The AWS account is preconfigured with required permissions for Lambda, S3, API Gateway, CloudFront, Route 53, Secrets Manager, and EventBridge.
- GitHub repository credentials are available and securely integrated with AWS CodePipeline and CodeBuild.
- All components will be deployed within a single AWS region to simplify architecture and reduce latency.
- The system will rely on AWS-managed services only, with no on-premises dependencies.
- Internet connectivity is available for Lambda functions to access external APIs.
- Project risks include temporary unavailability of third-party data sources or rate limits on external APIs.
- Security and reliability will adhere to the **AWS Well-Architected Framework** guidelines.

2 SOLUTION ARCHITECTURE / ARCHITECTURAL DIAGRAM

2.1 TECHNICAL ARCHITECTURE DIAGRAM

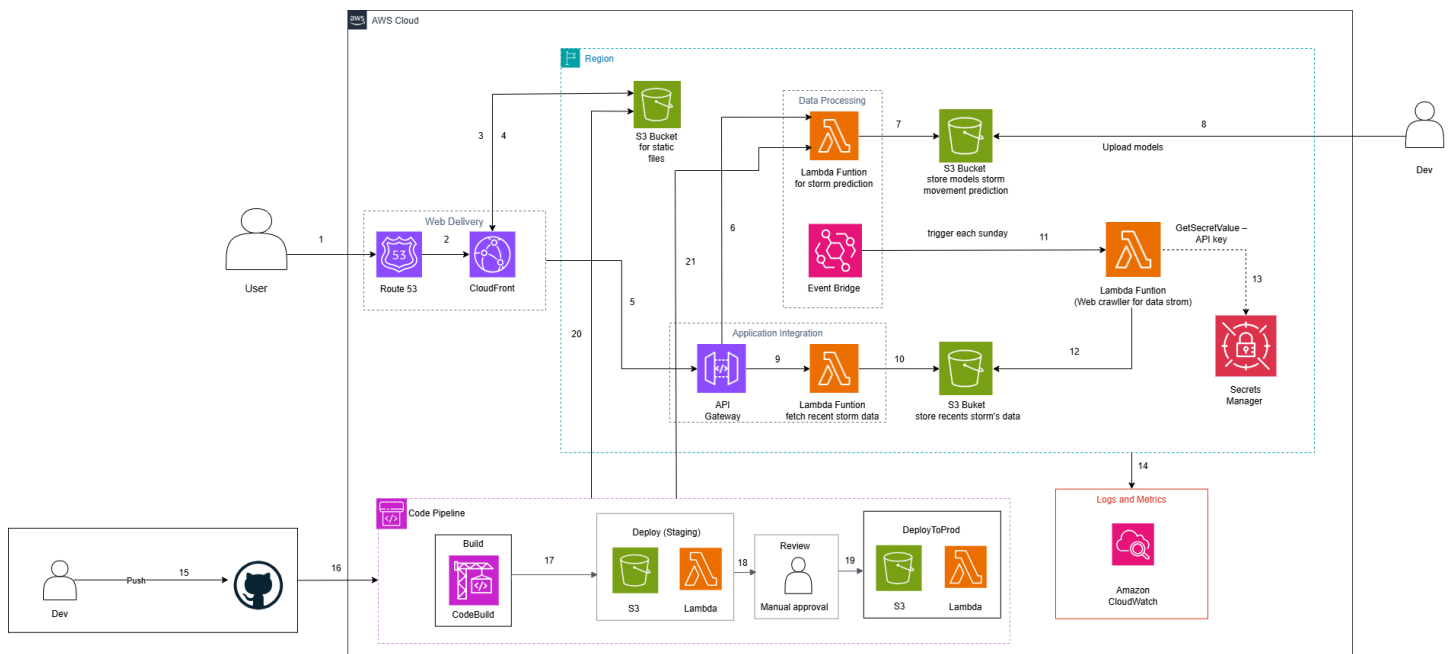
The proposed solution adopts a fully serverless, cloud-native architecture designed in alignment with the AWS Well-Architected Framework. The architecture focuses on scalability, security, cost-efficiency, and operational excellence. The system operates as a real-time hurricane trajectory prediction and visualization platform that integrates data ingestion, processing, model inference, and frontend delivery through managed AWS services.

Incoming requests from users are routed through Amazon Route 53 to Amazon CloudFront, which serves as a global content delivery network for static web assets hosted on Amazon S3. The web interface provides visualization dashboards and interfaces for model predictions.

Prediction requests and data retrievals are handled through Amazon API Gateway, which invokes AWS Lambda functions to process requests in a stateless and cost-effective manner. The Lambda functions load pre-trained models from Amazon S3 to perform inference, fetch updated storm data, or trigger the web crawler for new meteorological data. The automation pipeline is orchestrated by Amazon EventBridge, which schedules weekly or daily Lambda executions for data collection.

Processed data and model outputs are stored in Amazon S3, which serves as the central data lake. Logs, metrics, and operational traces are sent to Amazon CloudWatch for monitoring, performance tracking, and alert management. Secrets and API keys for external data sources are securely managed using AWS Secrets Manager. IAM roles enforce least-privilege access across all components, and AWS Certificate Manager provides TLS/SSL encryption for HTTPS endpoints.

The architecture adheres to the five pillars of the AWS Well-Architected Framework—operational excellence, security, reliability, performance efficiency, and cost optimization—ensuring a scalable and maintainable system suitable for both research and production environments.



List of Tools and Technologies

- AWS Lambda for serverless computing and model inference
- Amazon S3 for storage of datasets, models, and frontend assets
- Amazon API Gateway for RESTful API management
- Amazon CloudFront for global content delivery
- Amazon Route 53 for domain name routing
- Amazon EventBridge for task scheduling and automation
- AWS Secrets Manager for secure credential management
- Amazon CloudWatch for logging, metrics, and monitoring
- AWS IAM for access management and security control
- AWS Certificate Manager (ACM) for HTTPS/TLS certificates
- Next.js for frontend development
- Python (with boto3, TensorFlow/PyTorch) for backend and ML model integration

2.2 TECHNICAL PLAN

The development team will implement the infrastructure and deployment scripts using AWS CloudFormation and Terraform. This approach ensures consistent, repeatable, and automated provisioning of all AWS resources. Lambda functions will be developed in Python, packaged, and deployed through AWS CodePipeline and CodeBuild for continuous integration and delivery. Configuration changes, such as model version updates or API key rotations, will follow a controlled approval workflow with version tracking in AWS CodeCommit. Comprehensive test coverage will include functional, integration, and performance testing for all major components. CloudWatch alarms and dashboards will be configured to provide ongoing visibility into system performance and failures.

2.3 PROJECT PLAN

The project will adopt the Agile Scrum framework over eight one-week sprints. Each sprint will include planning, development, testing, and review phases. Stakeholders from the data science, cloud infrastructure, and application development teams will participate in sprint reviews and retrospectives. Communication cadence will include daily stand-ups, weekly progress reviews, and bi-weekly sprint demonstrations. Knowledge transfer sessions will be conducted near project completion to ensure operational readiness and team self-sufficiency.

2.4 SECURITY CONSIDERATIONS

The hurricane prediction platform applies AWS security best practices across all operational layers to ensure data protection, controlled access, and system reliability.

A. Access Security

- Enforce least-privilege IAM policies for all AWS resources so that each Lambda function and component only performs authorized operations.
- Apply role-based access for development, deployment, and data management to avoid permission overlap.
- Rotate IAM access keys periodically and remove unused credentials to maintain account integrity.

B. Infrastructure Security

- All public endpoints (frontend and APIs) are secured with HTTPS using certificates managed by AWS Certificate Manager.
- CloudFront distributions are configured with Origin Access Control (OAC) to prevent direct public access to S3 buckets.
- API Gateway connects securely to Lambda using AWS-managed authentication and authorization.
- Each Lambda function runs with an IAM execution role that limits access to only the required services such as S3, Secrets Manager, and CloudWatch.

C. Data Security

- S3 buckets use default encryption (SSE-S3) for data at rest and enforce HTTPS/TLS 1.2+ for all data transfers.
- Sensitive credentials and external API keys are stored in AWS Secrets Manager, ensuring encrypted access by authorized functions.

D. Monitoring and Detection

- CloudWatch Logs capture all Lambda executions, including function duration, errors, and memory usage.
- Logs are retained for one month in S3 for traceability and troubleshooting.

E. Incident Management

- CloudWatch alarms automatically notify system maintainers via email when a function fails or performs abnormally.
- S3 versioning supports quick restoration of previous data or models during failure recovery.

This configuration ensures that the hurricane prediction platform remains secure, fault-tolerant, and compliant with AWS best practices while maintaining a lightweight and cost-efficient architecture.

3 ACTIVITIES AND DELIVERABLES

3.1 ACTIVITIES AND DELIVERABLES

[Provide project milestones with timeline and respective deliverables, corresponding to the items and activities described in the Scope of Work / Technical Project Plan section. Indicate plans on how to govern the project/ change management; communication plans; transition plans]

Project Phase	Timeline	Activities	Deliverables/Milestones	Total man-day
Assessment	Week 1- 2	Review papers & problem scope Study datasets (ERA5, IBTrACS) Study AWS services conceptually	Requirement summary Problem definition	[10 man-day]
Setup base infrastructure	Week 2-3	Learn AWS fundamentals Understand cyclone forecasting domain	AWS study notes Service overview document	[7 man-day]
Setup component 1	Week 4	Create S3 bucket structure (static files, models, recent storm data) Setup Secrets Manager for API keys	S3 buckets ready Secrets Manager configured	[10 man-day]
Setup component 2	Week 5	Develop Lambda functions (crawler, fetch recent storm data, model prediction) Setup EventBridge schedule Store processed storm data in S3	Lambda functions deployed EventBridge rule active	[15 man-day]
Testing & Golive	Week 6-7	Integrate Lambda with API Gateway Test predictions on recent storms in the East Sea Check logs & metrics using CloudWatch Prepare static files for CloudFront	Test report API integration validated	[8 man-day]
Handover	Week 8	Prepare documentation Review CI/CD flow (GitHub → CodePipeline → CodeBuild → Deploy to S3/Lambda) Conduct knowledge transfer	Handover package CI/CD overview	[8 man-day]

3.2 OUT OF SCOPE

- Any non-serverless compute services such as EC2, ECS, EKS, or on-premise servers.
- Any database services including RDS, Aurora, DynamoDB, or data warehousing platforms.
- Any real-time satellite image ingestion, video processing, or high-resolution geospatial data pipelines.
- Any integration with external meteorological agencies or official government forecast systems.
- Any advanced authentication or user identity management features (e.g., Cognito, SSO, RBAC).
- Multi-region deployment, failover architecture, and disaster recovery strategies.
- Advanced frontend UI/UX design beyond basic static web delivery through S3 + CloudFront.
- Big data analytics platforms such as Athena, EMR, Glue ETL, or business intelligence tools.
- Streaming or messaging systems such as Kinesis, Kafka, or event buses outside of EventBridge triggers.
- Large-scale production hardening including cost optimization, auto-scaling, or enterprise monitoring.

3.3 PATH TO PRODUCTION

The current POC covers only the essential steps needed to verify the storm-prediction workflow. It does not yet include several elements required for a stable production release. Before going live, the system will need stronger error handling, broader testing, better monitoring, and tuning to support consistent performance. Additional checks around security, data validation, and operational reliability will also be needed to ensure the setup can run safely at scale.

4 EXPECTED AWS COST BREAKDOWN BY SERVICES

Region: *ap-southeast-1* (Singapore)

A dedicated AWS Pricing Calculator estimate has been created for this project, reflecting all services defined in the system architecture. The calculation includes On-Demand and long-term (reserved or committed) pricing where applicable. The estimate also covers tooling and migration components used to automate deployment and manage the project lifecycle.

A. Frontend & Content Delivery

- Amazon S3 (Static Files): Hosts 5 GB of frontend assets (HTML, CSS, JS) and transfers 10 GB per month. Cost $\approx$ \$0.54/month.
- Amazon CloudFront: Distributes 50 GB of data globally with up to 1 million requests per month. Cost $\approx$ \$6.00/month.
- Amazon Route 53: Provides 1 hosted zone and 1 million DNS queries per month. Cost $\approx$ \$0.90/month.
- AWS Certificate Manager (ACM): Issues and manages HTTPS certificates. Free of charge.

Subtotal for Frontend & CDN: $\approx$ \$7.4/month

B. Backend (API and Model Inference)

- Amazon API Gateway: Manages 1 million HTTP requests per month at an average size of 1 MB per request. Cost $\approx$ \$2.50/month.
- AWS Lambda (Storm Prediction): Executes model inference around 1,000 times per day with 512 MB memory and 1 GB ephemeral storage for 5 seconds per run. Cost $\approx$ \$2.54/month.
- AWS Lambda (Fetch Recent Storm Data): Runs $\sim$ 20,000 times per day with 512 MB memory for 1 second per run to retrieve stored data. Cost $\approx$ \$0.00/month (free tier).

Subtotal for Backend: $\approx$ \$4.54/month

C. Automation & Data Crawling

- Amazon EventBridge: Schedules weekly or daily automated tasks for data updates. Free of charge.
- AWS Lambda (Web Crawler): Collects weather data from external APIs using 128 MB memory and 512 MB ephemeral storage, $\sim$ 30 seconds per execution. Cost $\approx$ \$0.00/month (free tier).
- AWS Secrets Manager: Stores up to 5 API credentials securely for external weather sources. Cost $\approx$ \$2.00/month.

Subtotal for Automation & Data Crawling: $\approx$ \$2.00/month

D. Monitoring & Logging

- Amazon CloudWatch Logs: Collects logs from Lambda executions and stores them for one month ($\sim$ 2 GB/month). Cost $\approx$ \$0.57/month.
- CloudWatch Metrics (Lambda Insights): Tracks runtime and error metrics for Lambda functions. Cost $\approx$ \$0.00/month (free tier).

Subtotal for Monitoring & Logging: $\approx$ \$0.57/month

E. Storage & Data Transfer

- Amazon S3 (Model Bucket): Stores ML models (~1 GB) with ~60,000 GET requests per month. Cost ≈ \$0.05/month.
- Amazon S3 (Recent Storms Bucket): Stores recent storm data (~1 GB) with ~60,000 GET and 30 PUT requests per month. Cost ≈ \$0.27/month.

Subtotal for Storage & Data Transfer: ≈ \$0.32/month

F. Tooling & Migration

- AWS CodePipeline and CodeBuild for 10 minutes fixing ≈ \$0.90/month.

Total Estimated Monthly Cost

- Frontend & CDN: \$7.44
- Backend (API + ML): \$4.54
- Automation (Crawler + Secrets): \$2.00
- Monitoring & Logging: \$0.57
- Storage & Data Transfer: \$0.32
- Tooling & Migration: \$0.9

TOTAL MONTHLY COST : \$15.73/month

There is a mismatch compare to Price Calculator as some of the service ask to manually minus free tier plan from the total cost (e.g Cloudwatch metrics)

5 TEAM

Partner Executive Sponsor

Name	Title	Description	Email / Contact Info
Indiana Jones	Director of Weather & Climate Systems	Executive sponsor overseeing strategic alignment and storm-platform modernization	lucasfilm@gmail.com

Project Stakeholders

Name	Title	Stakeholder for	Email / Contact Info
Gale Weather	Head of Meteorological Analytics	Storm data modeling & analysis requirements	gale.505@gmail.com
Sidney Prescott	IT Operations Manager	Infrastructure reliability & data pipelines	mainscream@gmail.com

Partner Project Team

Name	Title	Role	Email / Contact Info
Minh-Duc Nguyen	Team Leader	Project Manager	minhducn8870@gmail.com
Quang-Duc Le Nguyen	Software Engineer	Fullstack, API Development	quangducle.127@gmail.com
Cong-Son Nguyen	AI Engineer	Data & Training Pipeline	yama27032005@gmail.com
Gia-Phat Nguyen	AI Engineer	Model Research, Evaluation	phat10102005@gmail.com
Tien-Dat Tran	Software Engineer	Backend service	dattt1608@gmail.com

Project Escalation Contacts

Name	Title	Role	Email / Contact Info
Minh-Duc Nguyen	Team Leader	Level 1 – Primary Escalation	dattt1608@gmail.com quangducle.127@gmail.com
Tien-Dat Tran	Software Engineer	Level 2 – Technical Escalation	
Quang-Duc Le Nguyen	Software Engineer	Level 3 – Final Escalation	

6 RESOURCES & COST ESTIMATES

Resource	Responsibility	Rate (USD) / Hour
Solution Architects [1]	Responsible for designing the serverless AWS architecture, defining Lambda workflows, S3 structure, security policies, and integration with ML models. [1]	50\$/Hour
Engineers [3]	Responsible for implementing Lambda functions, API Gateway integration, frontend setup on S3/CloudFront, automation scripts, ML model deployment, and testing. [3]	40\$/Hour
Project Manager	Oversees project planning, progress tracking, coordination between architecture and engineering teams, and ensures milestones are delivered on time. Handles documentation and stakeholder reporting. [1]	55\$/Hour

Project Phase	Solution Architects	Engineers	Project Manager	Total Hours
Design Architecture + Estimate Cost and Planning	20	15	15	50
Implement Serverless Infrastructure	10	90	10	110
Integrate ML Models & Automation	5	70	10	85
Test, Optimize, Deploy and Maintenance	5	45	25	75
Total Hours	40	220	60	320
Total Cost	2000\$	26400\$	3300\$	31700\$

Cost Contribution distribution between Partner, Customer, AWS:

Party	Contribution (USD)	% Contribution of Total
Customer	192020\$	60
Partner	9510\$	30
AWS	3170\$	10

7 ACCEPTANCE

Upon completion of each project phase, the Provider (“PROVIDER”) will submit the corresponding tangible and digital Deliverables to the Customer (“CUSTOMER”), accompanied by an official Acceptance Form as specified in Appendix B of this Statement of Work (SOW). Deliverables may include AWS architectural design documentation, Lambda source code, infrastructure configuration, data ingestion workflows, ML model integration outputs, testing reports, and any functional components of the online hurricane-education and visualization platform.

Following each submission, the CUSTOMER will have ten (10) business days (the “Acceptance Period”) to review, evaluate, and/or test the Deliverables to determine whether they meet the functional, technical, performance, and security acceptance criteria defined in this SOW. During the Acceptance Period, the CUSTOMER may verify that the Deliverables:

- Are deployed correctly within the AWS serverless architecture (including Lambda, API Gateway, S3, CloudFront, EventBridge, and Secrets Manager)
- Perform automated data ingestion, processing, and storm-trajectory prediction as described
- Display storm information and trajectory predictions correctly on the visual frontend
- Meet defined performance expectations such as latency, availability, and responsiveness
- Comply with the implemented security controls, IAM policies, and CloudWatch monitoring configurations
- Integrate correctly with supporting CI/CD processes (CodePipeline, CodeBuild)

If the CUSTOMER determines that a Deliverable satisfies all acceptance criteria in all material respects, the CUSTOMER will issue a written acceptance confirmation to the PROVIDER via the Acceptance Form before the end of the Acceptance Period.

If the CUSTOMER determines that a Deliverable does not meet one or more acceptance criteria, the CUSTOMER will issue a written Rejection Notice, including detailed descriptions of all defects, missing components, or non-conformities. The Rejection Notice must be submitted within the Acceptance Period.

Upon receiving a Rejection Notice, the PROVIDER will promptly correct the identified issues to ensure the Deliverable fully meets the SOW requirements and acceptance criteria. The corrected Deliverable will then be resubmitted to the CUSTOMER along with a new Acceptance Form.

For any resubmitted Deliverable, the CUSTOMER’s review will be limited to confirming:

1. Whether the PROVIDER has corrected the defects or non-conformities identified in the Rejection Notice, and
2. Whether the corrections have caused any unintended impact on related components or integrations.

If the CUSTOMER does not issue a Rejection Notice before the end of the Acceptance Period, the corresponding Deliverable will be considered deemed accepted.

Once all Deliverables for all project phases—including architecture design, serverless implementation, data automation, ML model integration, system testing, and final deployment—have been accepted, the CUSTOMER will provide a written Final Acceptance confirming successful completion of the project.